

Giant Vegetables & Fruits

Geographic Analysis



By Ellann Cohen & Carolina Pinillos
DATA 503: Data Engineering - Spring 2025

Motivation

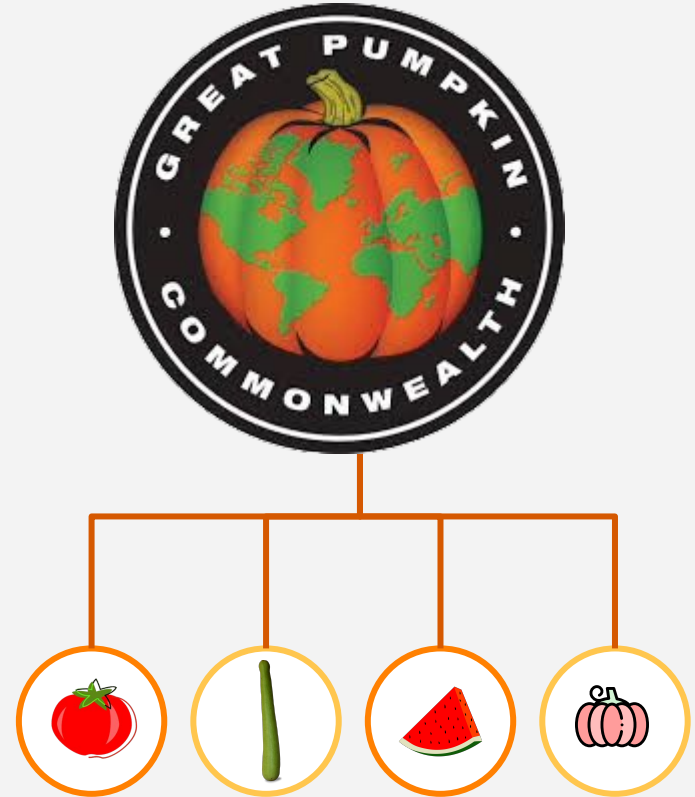
- Giant fruits and vegetables are amazing!
 - For example, the world record giant pumpkin weighed 2749 lbs!
- As a pumpkin community, we'd love to break the 3000 lb mark.
- Let's gather some data to see how we can help the community achieve this goal.

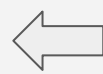


*Largest pumpkin grown in Oregon in 2024,
Jim Sherwood, 2453 lbs*

Background

- Great Pumpkin Commonwealth (GPC) is governing international organization that maintains rules and records around these giant fruits.
- They currently recognize and govern 8 categories of giant fruit





Atlantic Giant Pumpkin (AG)

WR: 2749 lbs

Bushel Gourd (BG)

WR: 470.5 lbs



Giant Squash (S)

WR: 2164 lbs

Watermelon (W)

WR: 350.5 lbs



*Images are from local PGVG growers, not the world record fruits.



Marrow (M)
WR: 253.5 lbs



Tomato (T)
WR: 11.65 lbs



Field Pumpkin (FP)
WR: 235 lbs

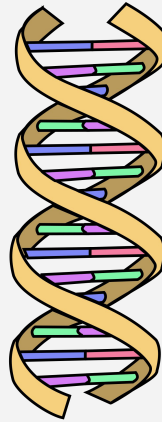
Long Gourd (LG)
WR: 173.75"



*Images are from local PGVG growers, not the world record fruits.

Factors that impact final fruit size

- Genetics (aka the seed)
- Location - macro & micro
- Soil
- Watering amount/frequency/quality
- Fertilizer
- Pest/disease management
- Temperature control (bury heating cables? greenhouse? shade cloth?)
- Mechanical care techniques
 - To bury or not bury vines, vine growth pattern, where to set fruit on vine, number of fruits grown per vine, etc
- Physical protection (if needed)
 - Animal, wind, hail protection
- Season specific weather conditions

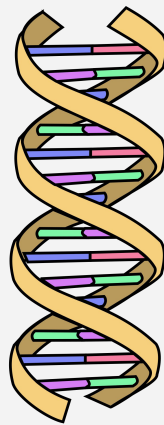


N P K



Factors that impact final fruit size

- Genetics (aka the seed)
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- Soil
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Research Top Question

What are the optimum locations (geographically) to grow different giant vegetables & fruits?

- Does it change by type of vegetable?

Between 40° and 50° latitude?

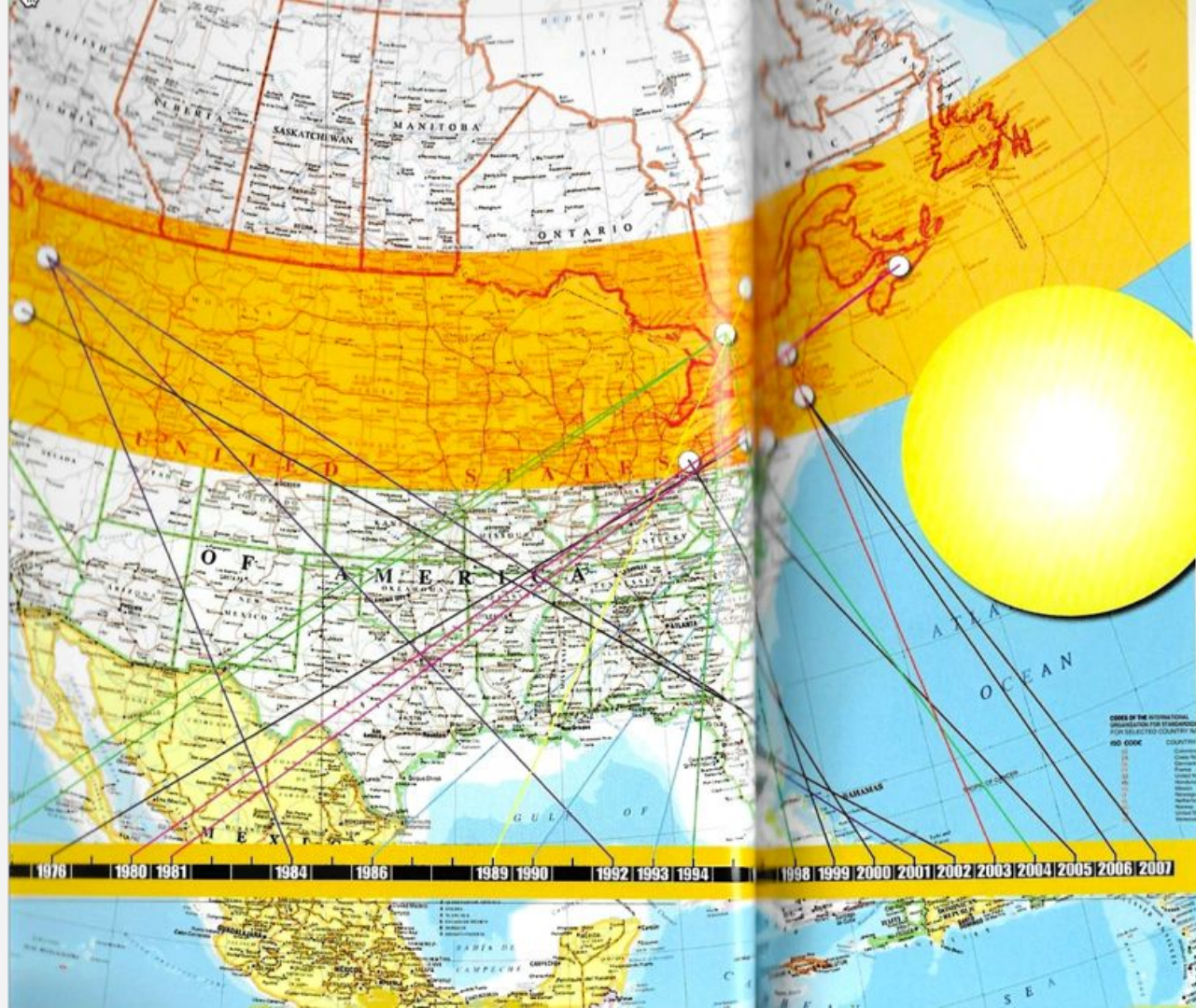
Donald Langevin claims that the optimal growing location is between 40° and 50° latitude in his book (published 2009).

Does this still hold in 2025?

The history, the people,
the pumpkins, the records

HOW-TO-GROW WORLD CLASS GIANT PUMPKINS THE ALL-ORGANIC WAY





kins in North America are represented by the orange-colored area stretching east and west from the Atlantic to the Pacific Oceans, and north and south from the 50th to the 40th latitudes. We could actually tweak this even further to that area between the 40th and 46th latitudes. All the world records have been grown within this "Orange Zone."

This zone suggests that the length of sunlight days (which is greater the further North you go in Summer) is important, and is only limited by the number of frost-free days (which diminish the further North you go). The happy medium is between the 40th and 50th latitudes where temperate summertime temperatures and adequate daylight combine to make growing conditions optimum.

Sun

I cannot stress enough the importance of sunshine, and the amount of it that your growing area directly receives each day. Pumpkins are day-neutral – meaning they mature and flower regardless of the length of their exposure to alternate periods of darkness and light; however, the more light they have, the greater the potential to manufacture carbohydrates by way of photosynthesis. That which the plant manufactures during sunlight is typically stored during darkness. It has been proven by various experiments using digital scales to monitor weight gain that gains are more pronounced during darkness. This is a common observational fact which reveals itself to any grower if measures more than once a day for estimation.

Area where giant pumpkins can be grown

Optimum growth areas

weight. During sunlight, the plant manufactures, and during darkness it stores. The more sunlight, the more the potential for weight gain.

Size of Growing Area

The size of the growing area has been a highly contested issue over the years. In the 1980s and 1990s, plant sizes of 3000 square feet or more were advocated, requiring areas of 60' x 50' for individual plants. However, in Warnock's, *How to Grow Big Squash*, he planted hills just 20' apart making an area of perhaps 500 square feet (20' x 25'). He was not specific about pruning techniques but he did "nip the ends of vines"

after fruit set. This would have made it possible to grow plants 20' apart at his late start, "about the eighteenth of May," and late pollinations "about the last week in July," would suggest side vine growth of 10' or less. Similar methods have become popular today with average plant sizes ranging from 500 to 1000 square feet – which can be accomplished by spacing plants 20' -30' apart. Given the rapid vegetative growth of the Atlantic Giant, even these seemingly large areas would be overtaken by criss-crossing vines if not for a determined effort to prune the plant. Pruning is essential to managing the plant for optimum fruit output, and will be covered in the chapter on *Vine Burying and Pruning*.

Soil Type

Optimum soil is best described as sandy loam containing silt and clay in moderate amounts, with humus. Too much silt or clay will reduce drainage resulting in less rejuvenation of air, and higher incidence of anaerobic bacteria (those which can secrete materials that are detrimental to root growth). Beware that even good sandy loam can have its drainage quality compromised by compaction, and compaction will stymie root and mycorrhizal growth.

How-to Test Your Soil's Drainage

Dig a hole about 1' deep. Fill with water and allow it to drain completely. Immediately dig the hole and measure the depth of the water in inches, and multiply by 4 to calculate how much water drains in an hour.

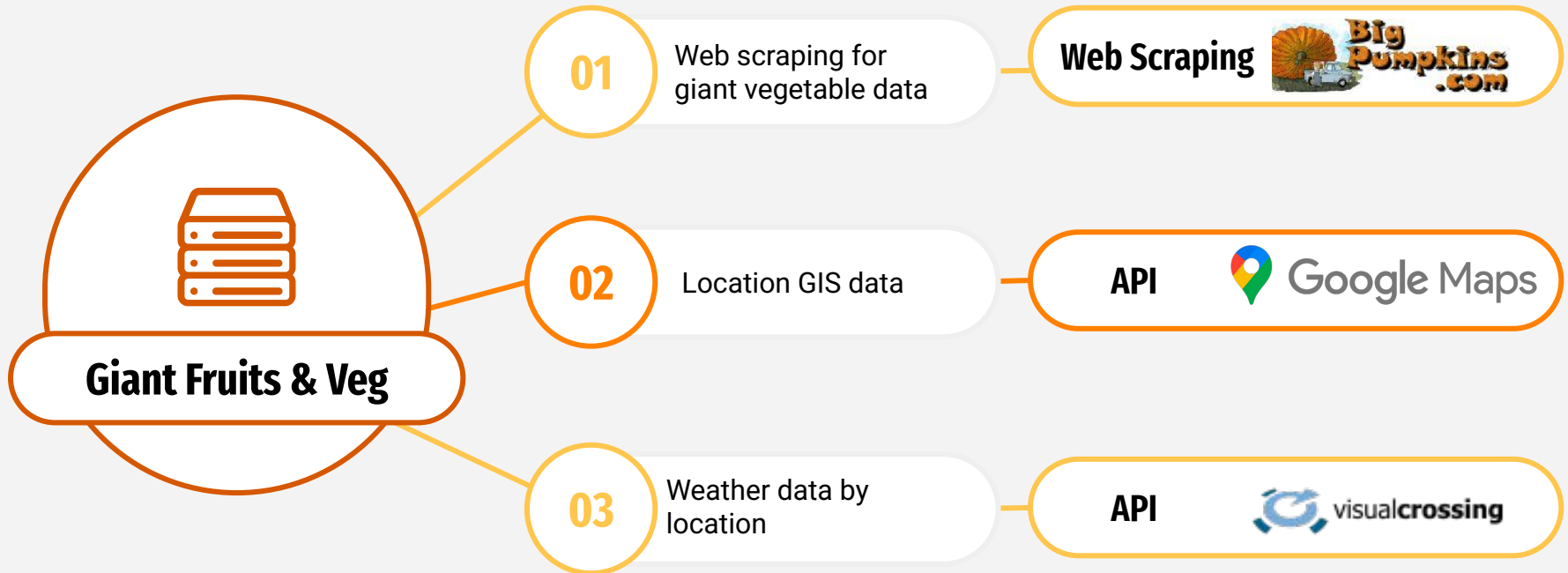
Less than 1" per hour is poor drainage. 1" per hour is normal for most soils. No more than 6" per hour may need additional organic amendments.

Data pipeline overview



Data Collection Methods

(Ingestion Phase)



Data ingestion - BigPumpkins

<http://www.bigpumpkins.com/WeighoffResultsGPC.aspx?c=P&y=2024>



GPC Weigh-off Results 2024: Pumpkin

[Home](#) | [What's New](#) | [Message Board](#) | [User Settings](#)

Select Destination

Site Search

This page shows the GPC results for the selected year and class. The results are not official until the GPC steering committee verifies them.



Place	Weight (lbs)	Grower Name	City	State/Prov	Country	GPC Site	Seed (Mother)	Pollinator (Father)	OTT	Est. Weight	Pct. Chart
1	2,731.50	Stelts, Dave & Carol	Enon Valley	Pennsylvania	United States	Ohio Valley Giant Pumpkin Growers Weigh-off	2144 Barron	2143 Stelts '23	491.0	2,312.00	18.0
2	2,670.00	Thompson, Earl	Rockford	Michigan	United States	Ohio Valley Giant Pumpkin Growers Weigh-off	2465.5 Sperry	2560 Gienger	495.0	2,353.00	13.0
3	2,641.10	Ian & Stuart Paton	Lymington	England	United Kingdom	Canna UK National Giant Vegetables Championship	2020 PATON	1727 JACOBUS	509.0	2,503.00	6.0
4	2,639.40	Paton, Ian & Stuart	Lymington	England	United Kingdom	Wargrave Nursery Giant Pumpkin Weigh-off	1727 Jacobus	2907 Paton	513.0	2,545.00	4.0
5	2,557.40	Van Geel, Mario	Kasterlee	Antwerp	Belgium	Belgisch Kampioenschap pompoenwegen	2560Gienger	2209BoonenHerman	488.0	2,284.00	12.0
6	2,551.00	Ford, Jim	Eldorado	Wisconsin	United States	River Prairie Ginormous Pumpkin Festival	2177 Ford	2042 Kisamore	463.0	2,030.00	26.0

Data ingestion - BigPumpkins

<http://www.bigpumpkins.com/WeighoffResultsGPC.aspx?c=P&y=2024>

Iterate over this letter
to choose different
category of fruit

Iterate over this
year to get all the
years of data

Scrapped in R:

- library(htmtools) # for saving html files
 - saved all html files
- library(rvest) # for scraping html files
 - looped thru html files to pull data
- Saved as one large csv file (~60k rows)

```
<table class="ReportResults" border="0">
  <thead>
    <tr>
      <th width="70">Place</th>
      <th>Weight (lbs)</th>
      <th>Grower Name</th>
      <th>City</th>
      <th>State/Prov</th>
      <th>Country</th>
      <th>GPC Site</th>
      <th>Seed (Mother)</th>
      <th>Pollinator (Father)</th>
      <th>OTT</th>
      <th>Est. Weight</th>
      <th>Pct. Chart</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td align="right">1</td>
      <td align="right">2,731.50</td>
      <td>Stelts, Dave & Carol</td>
      <td>Enon Valley</td>
      <td>Pennsylvania</td>
      <td>United States</td>
      <td>Ohio Valley Giant Pumpkin Growers Weigh-off</td>
      <td>2144 Barron</td>
      <td>2143 Stelts '23</td>
      <td align="right">491.0</td>
      <td align="right">2,312.00</td>
      <td align="right">18.0</td>
    </tr>
    <tr> ... </tr>
    <tr> ... </tr>
    <tr> ... </tr>
```

Data ingestion - Location

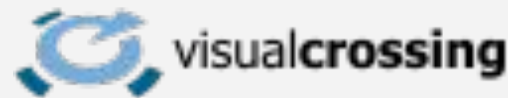


Location data

1. In R, pulled unique combinations of City, State/Prov, Country (saved as csv) (~ 5700 places)
2. In python, looped over the unique locations and fed to into the Google Maps API. (Ack: Prof. Cordova for helping with using the API.)
3. Saved latitude and longitude results into a new csv

City	State/Prov	Country	location	latitude	longitude
Northern Cambria	Pennsylvania	United States	Northern Cambria Pennsylvania United States	40.65923	-78.78169
Coventry	Rhode Island	United States	Coventry Rhode Island United States	41.68860	-71.56421
Little Valley	New York	United States	Little Valley New York United States	42.25256	-78.80559
Huntsburg	Ohio	United States	Huntsburg Ohio United States	41.53478	-81.05176
Sharon	Massachusetts	United States	Sharon Massachusetts United States	42.12367	-71.17895
Saegertown	Pennsylvania	United States	Saegertown Pennsylvania United States	41.71894	-80.14756
St. Catherines	Ontario	Canada	St. Catherines Ontario Canada	43.15937	-79.24686
Newbury	New Hampshire	United States	Newbury New Hampshire United States	43.32146	-72.03592
Johnston	Rhode Island	United States	Johnston Rhode Island United States	41.82052	-71.51262

Data ingestion - Weather



Weather data

1. In R, pulled unique combinations of City, State/Prov, Country (saved as csv) & year.
2. Joined with the latitude/longitude data, exported as csv
3. In python, looped over the unique latitude/longitude and year combinations to feed into the Visual Crossing API (Ack: Prof. Cordova for helping with using the API.)
4. Saved resulting weather data for all locations and years into a new csv. (~8 hrs runtime to pull data)

date	temp	tempmax	tempmin	dew	humidity	precip	precipprob	precipcover	pressure	cloudcover	visibility	sunrise	sunset	moonphase	windspeed	latitude	longitude
2005-01-01	47.2	52.8	36.8	39.6	75.3	0.000	0	0.00	1028.2	59.0	9.4	07:39:11	16:58:45	0.70	13.9	40.65923	-78.7
2005-01-02	42.6	49.5	35.8	36.4	78.9	0.069	100	20.83	1029.1	88.7	9.4	07:39:16	16:59:37	0.74	12.0	40.65923	-78.7
2005-01-03	49.8	53.2	44.2	48.8	96.4	0.750	100	87.50	1023.0	99.7	4.0	07:39:19	17:00:30	0.75	13.0	40.65923	-78.7
2005-01-04	44.4	49.9	40.8	42.3	92.3	0.119	100	37.50	1021.4	99.9	7.4	07:39:19	17:01:24	0.81	15.5	40.65923	-78.7
2005-01-05	35.0	40.6	30.4	33.8	95.3	1.066	100	100.00	1017.0	100.0	5.6	07:39:17	17:02:20	0.84	13.0	40.65923	-78.7
2005-01-06	35.1	45.6	30.9	32.9	91.6	1.089	100	91.67	1007.7	99.5	7.1	07:39:13	17:03:18	0.87	20.6	40.65923	-78.7

Data Collection Methods

(Transformation & Serving Phase)



Data Transformation into Database

Each sources .csv was loaded as entity in our postgres database.

city, state_prov, country -> keys

Hosted by Railway



Railway

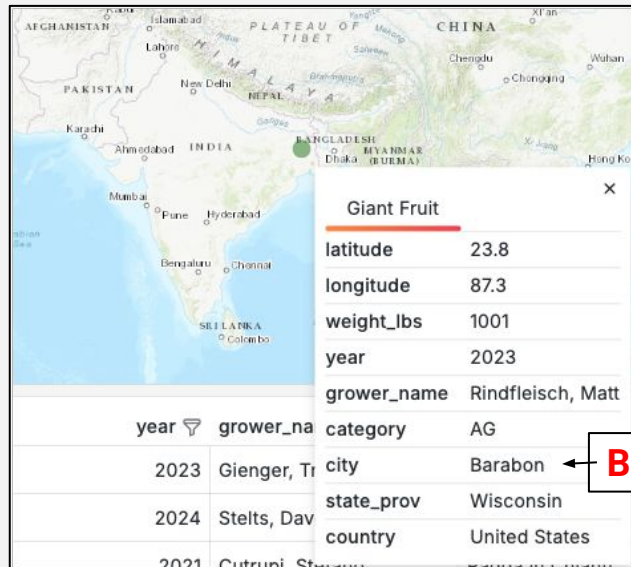
public
vegetables
place character varying
weight_lbs double precision
grower_name text
city text
state_prov text
country text
gpc_site text
seed_mother text
pollinator_father text
ott double precision
est_weight double precision
pct_chart double precision
category text
year integer
variety text

public
locations
id serial
city text
state_prov text
country text
location text
latitude double precision
longitude double precision

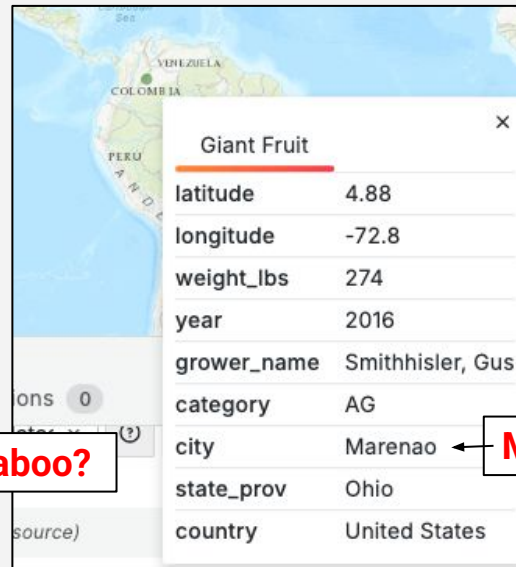
public
weather_daily
id serial
date date
temp double precision
tempmax double precision
tempmin double precision
dew double precision
humidity double precision
precip double precision
precipprob double precision
precipcover double precision
pressure double precision
cloudcover double precision
visibility double precision
sunrise time without time zone
sunset time without time zone
moonphase double precision
windspeed double precision
latitude double precision
longitude double precision

Data Cleaning - Some examples, user input errors

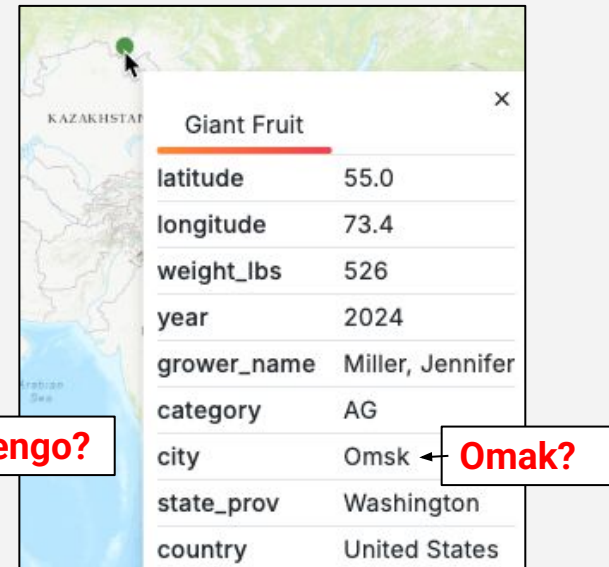
- WR Tomato: 11.65 lbs, yet several > 20 lbs in 2005 (data input error?)
- WR Long Gourd: 173.75", yet several > 300 in Europe (likely cm input?)
- Lat/Lon pulled incorrectly with typos
- Most likely haven't caught all data errors



Baraboo?



Marengo?



Omak?

Dashboard (Grafana)

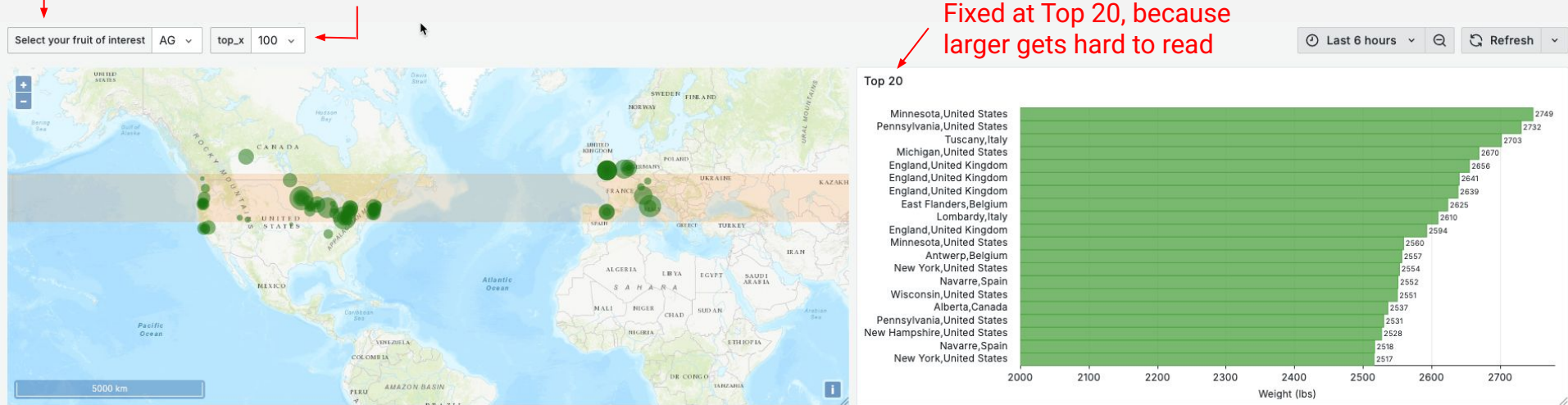
SQL queries with variables from dropdowns

Dropdowns:

1. Category of fruits and vegetables
2. Top 10, 20, 50, 100, 1000, etc

<https://grafana-ufa-production.up.railway.app/goto/IALo1pJHg?orgId=1>

Fixed at Top 20, because
larger gets hard to read

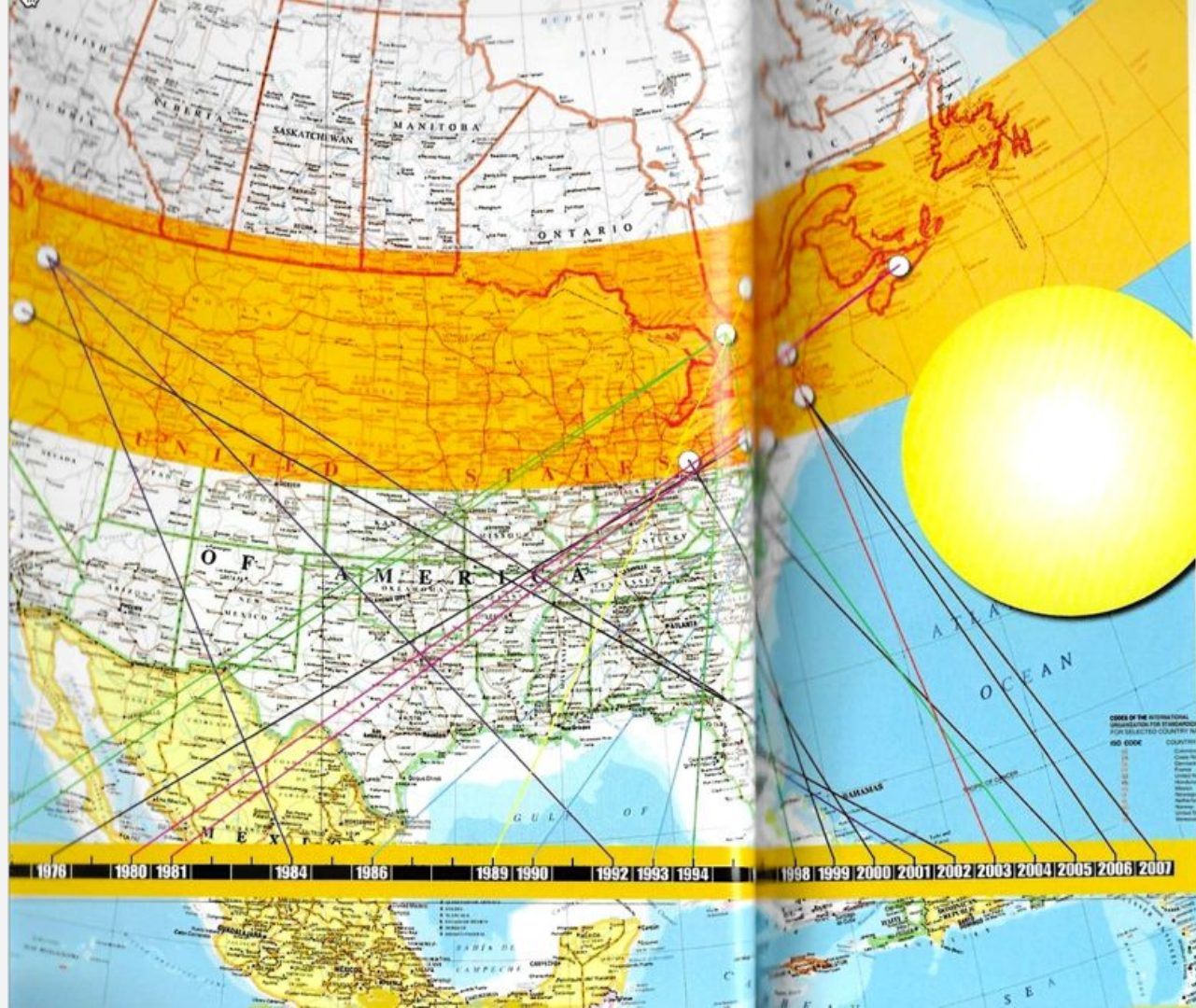


weight_lbs	year	grower_name	city	state_prov	country	seed_mother	pollinator_father	ott	pct_chart
2749	2023	Gienger, Travis	Anoka	Minnesota	United States	2365 Wolf	Sib	500	14
2732	2024	Stelts, Dave & Carol	Enon Valley	Pennsylvania	United States	2144 Barron	2143 Stelts '23	491	18
2703	2021	Cutrupi, Stefano	Radda in Chianti	Tuscany	Italy	1885.5 Werner	Self	514	6
2670	2024	Thompson, Earl	Rockford	Michigan	United States	2465.5 Sperry	2560 Gienger	495	13
2656	2022	Paton, Ian & Stuart	Lymington	England	United Kingdom	1940 Urena	2356.7 Mendi	497	12

Back to the key questions

What are the optimum locations (geographically) to grow different giant vegetables & fruits?

- Does it change by type of vegetable?



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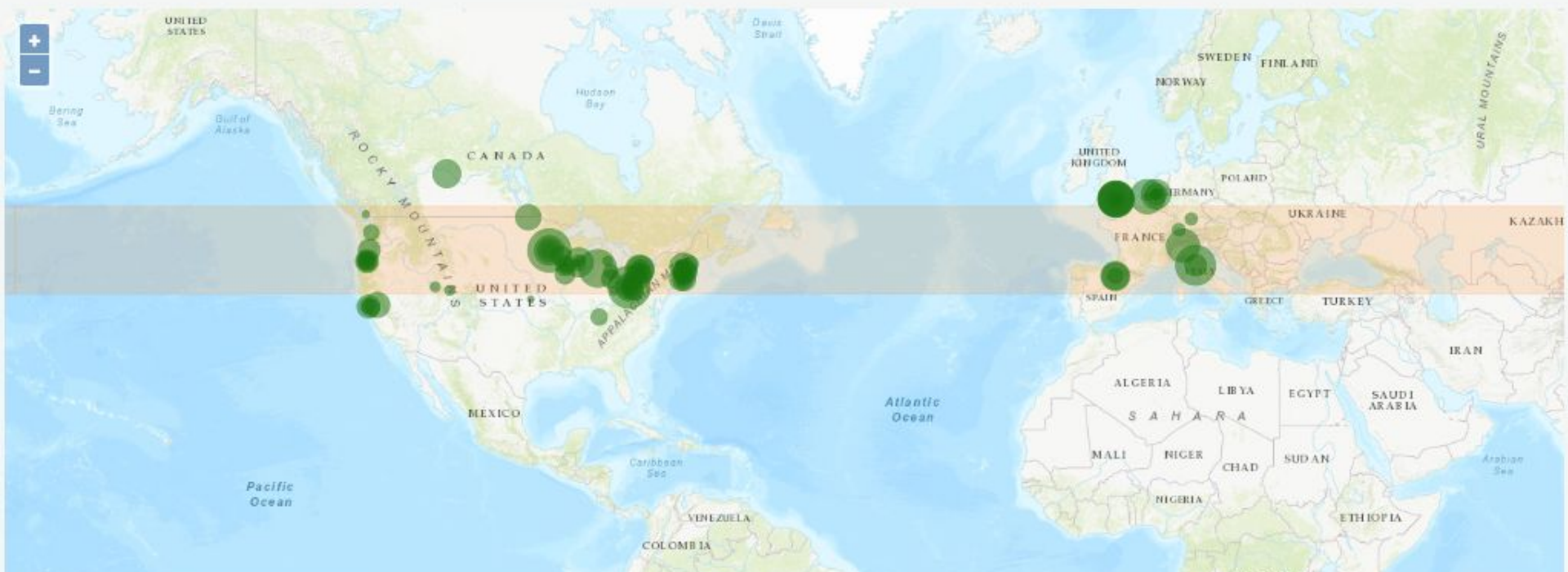
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For Atlantic Giant Pumpkins, Top 100

Donald Langevin's analysis still seems to hold. The top 100 AG's are a little beyond the bounds he initially suggested, but not too far off.

Select your fruit of interest AG top_x 100



Area where giant pumpkins can be grown

Optimum growth areas

weight. During sunlight, the plant manufactures, and during darkness it stores. The more sunlight, the more the potential for weight gain.

Size of Growing Area

The size of the growing area has been a highly contested issue over the years. In the 1980s and 1990s, plant sizes of 3000 square feet or more were advocated, requiring areas of 60' x 50' for individual plants. However, in Warnock's, *How to Grow Big Squash*, he planted hills just 20' apart making an area of perhaps 500 square feet (20' x 25'). He was not specific about pruning techniques but he did "nip the ends of vines"

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How-to Test Your Soil's Drainage

Dig a hole about 1' deep. Fill with water and allow it to drain completely. Immediately dig the hole and measure the depth of the water in inches, and multiply by 4 to calculate how much water drains in an hour.

Less than 1" per hour is poor drainage. 1" per hour is normal for most soils. No more than 6" per hour may need additional organic amendments.

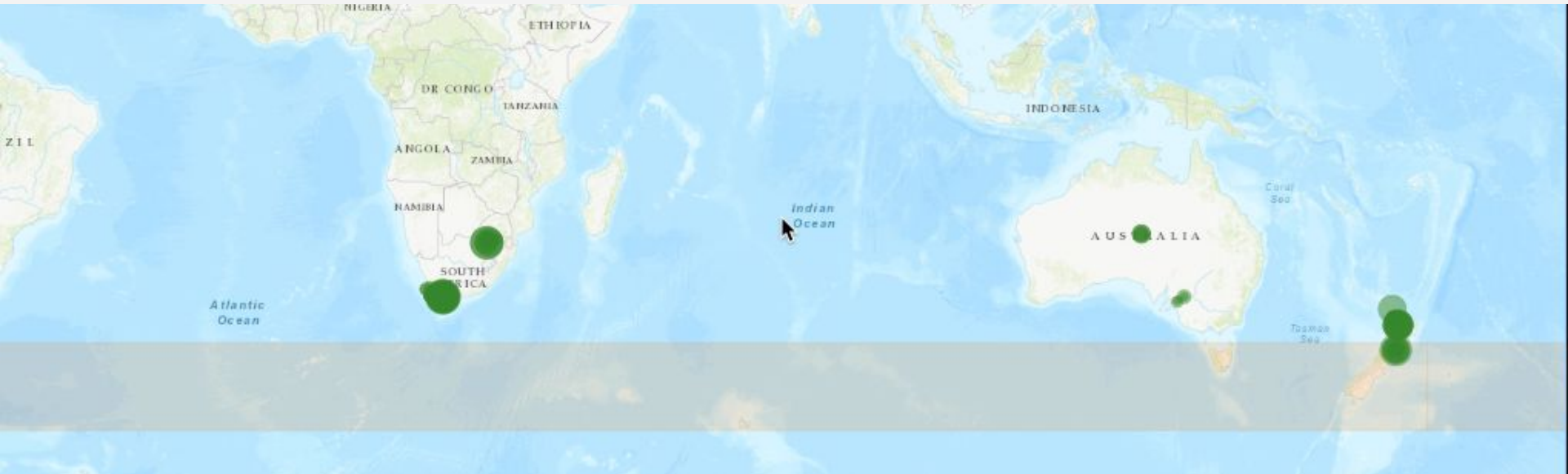
For Atlantic Giants, Top 500, focus on Southern Hemisphere

This band seems a little low and if Donald Langevin were to publish an updated version, we'd love to contribute and update his information with a new band.



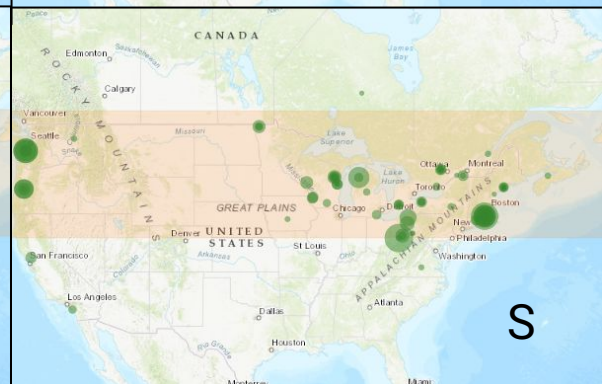
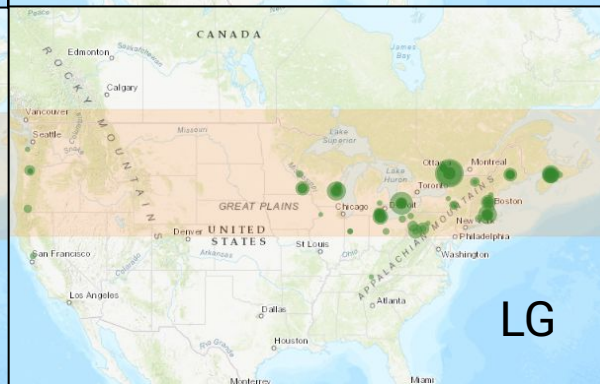
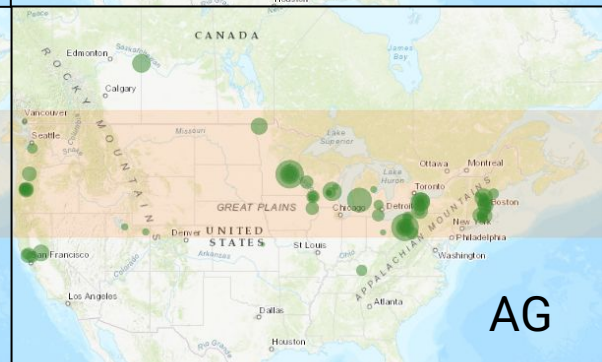
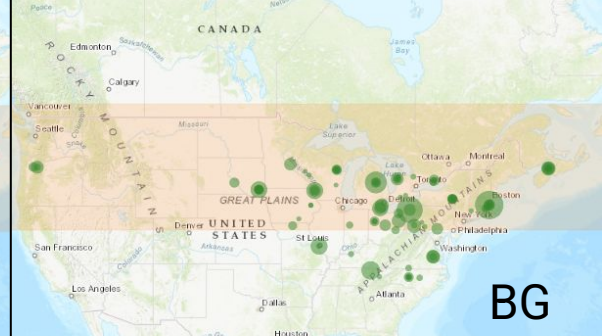
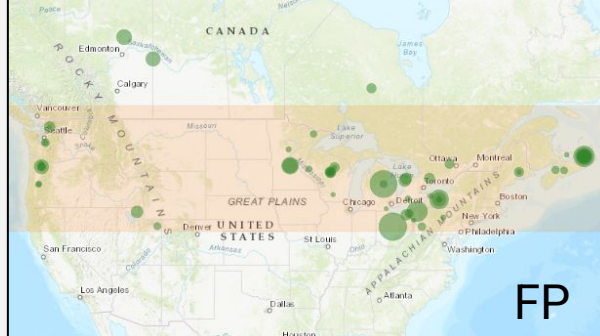
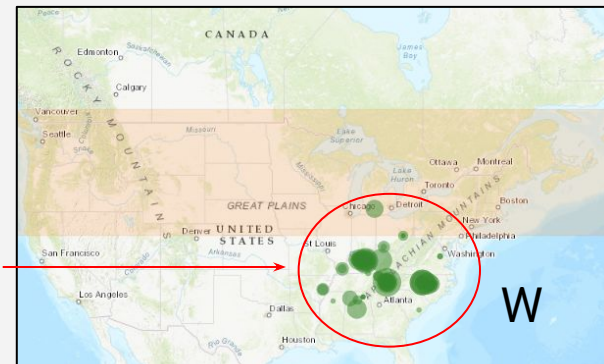
For Atlantic Giants, All, focus on Southern Hemisphere

Even when plotting all pumpkins, band still seems low.



Focus on N. America for all categories (Top 100 internationally)

Watermelons are an outlier.



Answers

What are the optimum locations (geographically) to grow different giant vegetables & fruits?

- *In general, the 40-50° lat. band is a good estimate of optimal growing range in the northern hemisphere. In Europe, the growth spills a little north of the band.*
- *In the southern hemisphere the optimal lat. band is north of the -40° lat. band.*

Does it change by type of vegetable?

- *Absolutely. Most categories share a band of optimal growth between 40-50° (within N. America) with a few outliers. **Watermelon** is the exception. All top watermelon (except for 1) have been grown south of 40° lat. line in N. America.*

Challenges, Lessons Learned, Future Work

- Weather data
 - It has been collected.
 - Too large a file size to upload to railway
 - Hard to connect and normalize into the database due to missing data and overlap between the two other tables.
 - Want to debug and analyze fruit growth with this data in the future.
 - Q: How to reduce daily weather data to represent the full growing season per fruit? TBD.
- Working with APIs
 - Often have strange limitations and want payment
 - only weather data for last calendar year? Why?
 - pulls lat/lon in USA but not Canada? Why?
- Dashboard tweaks and making it public
 - Add year data to the tool tip on the bar chart (without adding bars of the years)
 - Change the categories drop_down to words instead of acronyms
 - Grafana doesn't current support the dropdowns in public dashboards

Thank you!